

## Program2

### Develop a Multi Staged Dockerfile for container Orchestration

Program structure

```
|  
|---Dockerfile  
|---src/index.js  
|---package.json  
|---node_modules  
|---package-lock.json
```

#### Step1: Create the files

##### Dockerfile

---

```
FROM node:20-alpine AS builder  
WORKDIR /app  
  
# Copy everything first  
COPY . .  
  
# Install after copying  
RUN npm install  
  
# Build dist folder  
RUN npm run build  
  
# Second stage  
FROM node:20-alpine  
  
WORKDIR /app  
  
# Copy only what the final app needs  
COPY --from=builder /app/dist ./dist  
COPY --from=builder /app/node_modules ./node_modules  
COPY --from=builder /app/package.json ./package.json  
  
EXPOSE 3000  
CMD ["node", "dist/index.js"]
```

## Step2: build and run the container

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg2$ mkdir src
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg2$ nano index.js
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg2$ nano src/index.js
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg2$ nano package.json
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg2$ docker build -t program2 .
[+] Building 43.9s (14/14) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 474B                                0.0s
=> [internal] load metadata for docker.io/library/node:20-alpine  7.3s
=> [auth] library/node:pull token for registry-1.docker.io        0.0s
=> [internal] load .dockerignore                                    0.0s
=> => transferring context: 2B                                       0.0s
=> [internal] load build context                                    0.0s
=> => transferring context: 1.18kB                                   0.0s
=> [builder 1/5] FROM docker.io/library/node:20-alpine@sha256:658d0f63e501824d6c23e06 32.1s
=> => resolve docker.io/library/node:20-alpine@sha256:658d0f63e501824d6c23e06d4bb95c71 0.0s
=> => sha256:658d0f63e501824d6c23e06d4bb95c71e7d704537c9d9272f488ac03a 7.67kB / 7.67kB 0.0s
=> => sha256:fcbb8f7d018707c656a4da2eea8a15f2893d2258093fea9ff2ea5ea1c 1.72kB / 1.72kB 0.0s
=> => sha256:e80b0510ba947015cacddea3d23dc761e399971c074a1bf32eea7e4 6.52kB / 6.52kB 0.0s
=> => sha256:1074353eec0db2c1d81d5af2671e56e00cf5738486f5762609ea33d6 3.86MB / 3.86MB 11.1s
=> => sha256:8d06ba6946d1f299d8f962e37906c77006df33d47050430a57f789 42.78MB / 42.78MB 30.5s
=> => sha256:cb3325e64457574e24f92246e3da3376946e473d636209e1390eac47b 1.26MB / 1.26MB 3.2s
=> => sha256:fd1849a5c548bc65ee47a64498951bda5d40e87d08efe9dca69b5c8cdedc7 443B / 443B 4.1s
=> => extracting sha256:1074353eec0db2c1d81d5af2671e56e00cf5738486f5762609ea33d606f886 0.1s
=> => extracting sha256:8d06ba6946d1f299d8f962e37906c77006df33d47050430a57f7893ec35af6 1.3s
=> => extracting sha256:cb3325e64457574e24f92246e3da3376946e473d636209e1390eac47b50b26 0.0s
=> => extracting sha256:fd1849a5c548bc65ee47a64498951bda5d40e87d08efe9dca69b5c8cdedc7a 0.0s
=> [builder 2/5] WORKDIR /app                                       0.1s
=> [builder 3/5] COPY . .                                           0.0s
=> [builder 4/5] RUN npm install                                    3.2s
=> [builder 5/5] RUN npm run build                                  0.5s
=> [stage-1 3/5] COPY --from=builder /app/dist ./dist              0.0s
=> [stage-1 4/5] COPY --from=builder /app/node_modules ./node_modules 0.1s
=> [stage-1 5/5] COPY --from=builder /app/package.json ./package.json 0.1s
=> exporting to image                                               0.2s
=> => exporting layers                                              0.1s
=> => writing image sha256:1fbbbcd9835048d6f0fce81cce59705bb5e066c9f38b24221f536038044 0.0s
=> => naming to docker.io/library/program2                          0.0s
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg2$ docker run -it -p 10001:3000 program2
app is listening on PORT 3000
```

## Output



localhost:10001



For quick access, place your bookmarks here on the bookmarks bar. [Import bookmarks](#)

HERE S amulti stage docker file